

ARTIFACT TEMPLATE LIBRARY

Enterprise Architecture Lifecycle Artifacts

TOGAF 10 ADM · All Phases · AI-First Extensions

Every artifact shown as a live template with field definitions and example content

PRE
Preliminary

A
Vision

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Business

C1
Data

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Application

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Technology

E/F
Opportunities
& Migration

G
Governance

H
Change Mgmt

EA
Cross-Cutting
Artifacts

30+

Artifact Templates

10

ADM Phases Covered

100%

AI-First Extended

HOW TO USE THESE TEMPLATES

Blue italic text	<i>Example field values filled in for a fictional "GlobalBank AI Transformation Programme 2026". Replace with your actual data.</i>
Field labels	Every label is mandatory unless marked [OPTIONAL]. Mandatory fields require approval before phase advancement.
AI Extension fields	Fields marked ♦ are AI-First extensions added to TOGAF 10. These are NOT in standard TOGAF 10 but are required for AIDLC-integrated EA.
Artifact IDs	Prefix format: EA-[Phase]-[Type]-[NNN]. E.g. EA-B-HATB-001 = Business Architecture, Human-AI Task Boundary, first instance.
Linked artifacts	Each artifact shows upstream inputs (←) and downstream outputs (→) to trace lineage through the ADM cycle.

TOGAF 10 COMPLIANCE: All standard TOGAF 10 artifacts are included at their minimum viable form. AI-First extension fields (♦) are additive — they do not replace standard TOGAF artifacts. Organisations pursuing ISO/IEC 42001 or EU AI Act compliance should use AI-First fields as mandatory.

TOGAF ADM
Phase PRE

Preliminary

ARTIFACTS: EAPC-001 - ACP-001 - ESR-001

The Preliminary phase establishes the EA framework, governance model, and AI governance principles before any architecture work begins. In AI-first enterprises, this phase must produce a Constitutional AI commitment and establish the AI Governance Council as an EA stakeholder.

EA GOVERNANCE

EA Framework Charter

Owner: Chief Architect / CIO | ADM Phase: PRE | TOGAF 10 + AI-First Extension

CHARTER SCOPE & AUTHORITY

Organisation	GlobalBank plc — Digital & Technology Division
EA Framework Adopted	TOGAF 10 (The Open Group Architecture Standard, 10th Edition, 2022)
Charter Version / Status	v2.1 — Approved 15 January 2026
EA Programme Scope	Enterprise-wide: all business units, all technology investments >£500K, all AI systems (any cost)
EA Authority	Architecture Review Board (ARB) has binding authority over all technology architecture decisions. No AI system may be deployed without ARB sign-off at AIDLC Phase 6.
Chief Architect	Dr. Priya Mehta, Group Chief Architect (Board-level authority)
Review Cycle	Annual framework review; quarterly AI governance review; immediate review on regulatory change

◆ AI-FIRST EXTENSIONS

◆ AI Governance Council	Established as permanent EA stakeholder body. Composition: CRO, CPO, CISO, Chief Data Officer, Chief Architect, External AI Ethics Advisor. Quorum: 4 of 6 required for AI system approval.
◆ Constitutional AI Commitment	GlobalBank commits to AIDLC governance for all AI systems. Constitutional AI Policy is a mandatory output of Phase 4 (Model Design). No AI system proceeds to development without an approved CAP.
◆ AIDLC Integration	TOGAF ADM phases B, C, D incorporate AIDLC Phase Gates as mandatory Architecture Compliance milestones. See ACC-001 for the compliance checklist.
◆ Regulatory Alignment	EA framework explicitly aligned with: EU AI Act (all risk tiers), NIST AI RMF (Govern-Map-Measure-Manage), ISO/IEC 42001, FCA operational resilience requirements.

EA GOVERNANCE

AI Governance Principles

Owner: AI Governance Council | ADM Phase: PRE | TOGAF 10 + AI-First Extension

#	Principle	Statement	Implementation Mechanism
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AGP-1	Human Primacy	AI systems augment human judgment; they do not replace it for consequential decisions. Human override capability is mandatory in all TIER 1–2 AI systems.	HITL controls documented in AIDLC Phase 4; HITL trigger rates monitored Phase 8
AGP-2	Explainability by Design	Explainability is an architecture constraint, not a post-hoc addition. Every AI system must explain its outputs in terms meaningful to affected stakeholders.	Explainability approach selected in Phase 4 ADR; tested in Phase 6 red-teaming
AGP-3	Fairness & Non-Discrimination	AI systems must not perpetuate or amplify discrimination against legally protected groups. Bias assessment is mandatory before and after training.	Bias Baseline in Phase 3; Fairness Evaluation in Phase 6; ongoing monitoring Phase 8
AGP-4	Privacy by Design	Personal data used in AI must be processed lawfully, fairly, and with purpose limitation. PII must be masked before LLM processing.	Privacy Impact Assessment Phase 3; data minimisation in Data Sheet; PII masking in deployment
AGP-5	Transparency	Individuals affected by AI decisions must be informed that AI was used, understand the decision basis, and have a clear path to challenge.	User Disclosure Documentation Phase 7; adverse action notices; appeal mechanisms
AGP-6	Security & Robustness	AI systems must be designed to resist adversarial attack, distributional shift, and prompt injection. Security is assessed by the red team in Phase 6.	AI Threat Model Phase 4; Red Team Report Phase 6; Zero Trust enforcement Phase 7
AGP-7	Accountability	A named individual owns every deployed AI system. Ownership includes governance compliance, performance monitoring, and incident response.	AI System Inventory records owner; owner signs off Phase 7 deployment runbook
AGP-8	Proportionality	AI governance intensity scales with risk tier. TIER 4 (minimal risk) follows lightweight AIDLC; TIER 1 (unacceptable risk) is prohibited.	Risk Classification (RCS-001) determines AIDLC governance track at Phase 1

EA Governance

EA Stakeholder Register

Owner: Chief Architect | ADM Phase: PRE | TOGAF 10 + AI-First Extension

Stakeholder Group	Members	Interest	Influence	EA Role	Engagement Points
AI Governance Council	CRO, CPO, CISO, CDA, Chief Architect, Ethics Advisor	High	High	Approve AI system deployments; review CAP; set AGP	Monthly governance review; Phase 4 & 6 gates
Architecture Review Board (ARB)	Chief Architect, Domain Architects, Security Architect, Data Architect	High	High	Approve all architecture decisions; review ADRs	Phase gate reviews; emergency sessions for P0 incidents

Business Unit Owners	VPs per domain (Credit, Operations, Marketing, HR, Security)	High	Medium	Sponsor AI use cases; approve business architecture	Phase A discovery; Phase B capability mapping
Data Governance Board	Chief Data Officer, Domain Data Owners, DPO	High	High	Approve data strategy; sign off Data Sheet and PIA	Phase C1 data architecture reviews
◆ Model Risk Management	Chief Risk Officer, Model Risk Managers	High	High	Assess and sign off high-risk AI model deployments	Phase 6 evaluation sign-off; Phase 8 quarterly audit
◆ External Regulators	FCA, ICO, EU AI Act National Authority	High	Low (can influence)	Regulatory compliance; supervisory examination	Annual regulatory reporting; incident notification <72h
Technology Teams	Engineering, DevOps, MLOps, Data Engineering	Medium	High	Implement architecture; operate AI systems	All phases; primary AIDLC delivery
◆ Affected Customers	Retail and corporate banking customers	High	Low (consulted)	Fairness, transparency, right to explanation	FRIA consultation; user testing; appeal channel

TOGAF ADM
Phase A

Architecture Vision

ARTIFACTS: SAW-001 · ACA-001 · AVD-001

PHASE OUTPUT

Statement of Architecture Work

Owner: Chief Architect | ADM Phase: A | TOGAF 10 + AI-First Extension

WORK DEFINITION

Architecture Work Title	GlobalBank AI Transformation Programme — Phase 1: Core Banking AI Integration
Sponsoring Executive	Marcus Chen, Chief Operating Officer
Architecture Work Description	Design and govern the enterprise architecture for integrating AI systems into core banking operations: credit risk scoring, customer service automation, fraud detection. Encompasses 5 AI use cases (see ACA-001), full AIDLC lifecycle governance, and technology platform modernisation to support AI at scale.
Scope	IN SCOPE: UK Retail Banking division, core banking AI systems, data architecture for AI, MLOps platform, agent architecture. OUT OF SCOPE: International operations (Year 2), mortgage AI (separate programme), investment banking (separate governance).
Constraints	1. EU AI Act compliance required for all credit, HR and fraud systems. 2. FCA operational resilience rules apply. 3. Core banking system (Temenos) cannot be replaced in this programme. 4. Maximum cloud budget: £2.8M/year.
Assumptions	Alternative data partners (utility, rent) contracts will be signed by Q2 2026. Azure ML platform will remain enterprise standard for ML workloads. LLM provider contracts available under existing Microsoft EA.
Deliverables	Architecture Vision (AVD-001), Business Architecture (Phase B), Data Architecture (Phase C1), Application Architecture (Phase C2), Technology Architecture (Phase D), Migration Plan (Phase F), Governance Framework (Phase G)
Timeline	Phase A-D: Q1 2026 (12 weeks). Phase E-F: Q2 2026 (8 weeks). Phase G onwards: ongoing.
Approved By	Dr. Priya Mehta (Chief Architect) · Marcus Chen (COO) · AI Governance Council — 22 January 2026

✦ AI EXTENSION

AI Capability Assessment

Owner: Chief Architect + AI Lead | ADM Phase: A | TOGAF 10 + AI-First Extension

✦ **AI-First Extension:** This artifact is not in standard TOGAF 10. It provides the AI capability inventory and maturity baseline that drives all subsequent AI-impacted architecture decisions.

AI Use Case	EU AI Act Tier	AIDLC Phase	Model Type	Strategic Status	Priority	Compliance Requirements
Credit Risk AI Scoring	T2 — HIGH RISK	Phase 2 (Feasibility)	XGBoost + LLM	Foundational — must deliver	Priority 1	FRIA required; FCA regulated

Customer Service Copilot	T3 — LIMITED RISK	Phase 7 (Production)	GPT-4o RAG	Already live — extend scope	Priority 2	EU AI Act Art. 50 transparency
Fraud Detection AI	T2 — HIGH RISK	Phase 8 (Monitor)	Ensemble + LSTM	Mature — governance uplift	Priority 1	PCI-DSS; FCA SYSC 23
CV Screening Assistant	T2 — HIGH RISK	Phase 2 (HOLD)	LLM + Embedding	On hold — FRIA first	Priority 3	EU AI Act Annex III Cat.1
Campaign Targeting AI	T4 — MINIMAL RISK	Phase 5 (Dev)	XGBoost	Progressing normally	Priority 4	GDPR profiling consent
Developer Copilot (GitHub Copilot)	T4 — MINIMAL RISK	Phase 7 (Prod)	Codex/GPT-4o	Shadow AI — register & govern	Priority 3	AIDLC-Lite track

◆ AI Maturity Baseline	<i>McKinsey AI Maturity Level 2 of 5 (Developing). Proof-of-concept capability exists. Production MLOps immature. Governance framework nascent. Target: Level 3 (Scaling) by end 2026.</i>
◆ Shadow AI Exposure	<i>27 unregistered AI tools identified in SaaS survey (GitHub Copilot, Grammarly, Otter.ai, various ChatGPT integrations). All to be assessed and registered in AI System Inventory within 30 days.</i>
◆ Architecture Readiness	<i>Data architecture: MEDIUM (data lake exists, lineage immature). MLOps: LOW (no enterprise platform). Application: MEDIUM (microservices partially adopted). Security: LOW (no zero trust). Governance: LOW (nascent).</i>

PHASE OUTPUT

Architecture Vision Document

Owner: Chief Architect | ADM Phase: A | TOGAF 10 + AI-First Extension

Vision Statement	<i>By end 2026, GlobalBank will operate a trusted, explainable, and regulatorily compliant AI capability that delivers £7.3M+ annual value from credit risk automation, reduces manual review by 35%, and serves as a model for responsible AI in UK financial services.</i>
Target Architecture Descriptor	<i>AI-First Enterprise Architecture: 7-layer reference stack (L1 GPU Infrastructure → L7 Business & Governance), AIDLC lifecycle for all AI systems, Data Mesh for AI data products, Zero Trust for AI agents, TOGAF 10 ADM as the governance backbone.</i>
Key Architecture Principles	<i>1. Architecture before AI (no AI before the foundation is ready). 2. Data as first-class product. 3. Explainability by design. 4. Zero Trust at every layer. 5. Governance proportionate to risk. 6. Open standards over vendor lock-in.</i>
Business Outcomes	<i>60-second credit decisions (vs 72 hours), <5% false rejection rate (vs 12%), 100% adverse action explanation coverage, £7.3M ROI Year 1, ISO/IEC 42001 certification Q4 2026.</i>
Architecture Risks	<i>1. Legacy Temenos integration complexity (HIGH). 2. EU AI Act compliance timeline (HIGH). 3. Alternative data vendor contracts (MEDIUM). 4. MLOps talent gap (MEDIUM). 5. Shadow AI exposure (LOW-MEDIUM).</i>
ARB Approved	<i>5 February 2026 — Architecture Review Board unanimous approval</i>

TOGAF ADM
Phase B

Business Architecture

ARTIFACTS: ACM-001 - BATB-001 - AOM-001 - WIA-001

AI Extension

AI Capability Map

Owner: Enterprise Architect + Business Analysts | ADM Phase: B | TOGAF 10 + AI-First Extension

AI-First Extension: Extends the standard TOGAF Business Capability Map to include AI capabilities, their risk classification, AIDLC status, and Human-AI task boundary designation.

Business Capability	Domain	AI System	Risk Tier	Human-AI Boundary	AIDLC Status	Target Date
Credit Decision	Retail Banking	Credit Risk AI Scoring	T2 HIGH RISK	AI-Primary with HITL for <70% confidence cases	Phase 2	Phase 4 Q2 2026
Adverse Action Explanation	Retail Banking	LLM Adverse Action Generator	T2 HIGH RISK	AI-Primary (SHAP-grounded); human review on appeal	Phase 2	Phase 7 Q3 2026
Customer Enquiry Resolution	Operations	Customer Service Copilot	T3 LIMITED	AI-Augmented; human escalation for complex cases	Phase 7 (LIVE)	Scope extension Q2 2026
Fraud Transaction Detection	Security	Fraud Detection Ensemble	T2 HIGH RISK	AI-Primary; all alerts to human analyst	Phase 8 (LIVE)	Governance uplift Q2 2026
Candidate CV Screening	HR	CV Screening Assistant	T2 HIGH RISK	AI-Recommend ONLY; human decides	Phase 2 (HOLD)	Subject to FRIA outcome
Marketing Segmentation	Marketing	Campaign Targeting AI	T4 MINIMAL	AI-Primary; no individual impact	Phase 5	Phase 7 Q3 2026
Developer Code Assistance	IT/Engineering	GitHub Copilot (registered)	T4 MINIMAL	AI-Augmented; developer reviews all output	Phase 7 (LIVE)	Registered; AIDLC-Lite
Architecture Documentation	EA	EA AI Assistant (proposed)	T4 MINIMAL	AI-Augmented; architect validates	Phase 1	Proposed for Q3 2026

AI Extension

Human-AI Task Boundary Map

Owner: Business Architect + Ethics Lead | ADM Phase: B | TOGAF 10 + AI-First Extension

AI-First Extension: Documents the precise boundary between human and AI decision authority for every AI-impacted business process. Mandatory for all TIER 1–2 systems. This artifact is reviewed by the AI Governance Council and forms the basis for HITL design in AIDLC Phase 4.

Process Step	Human-Only Tasks	AI-Only Tasks	HITL Trigger & Human Role	Decision SLA
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Credit Application Intake	Capture applicant data; verify identity	AI validates data completeness and identity (automated)	Human if identity verification fails (manual override required)	Instant / 5 min
Initial Credit Score	N/A (fully automated for T4 decisions)	AI scores application (XGBoost model); assigns confidence	Human review for ALL applications <70% confidence (~8% of volume)	<60 seconds / 4 hours (HITL)
Adverse Action Notice	N/A	AI generates SHAP-grounded rejection notice (LLM)	Human verifies reason codes before delivery if flagged by SHAP-LLM mismatch check	2 min / 30 min (if flag)
Customer Appeal	Receive appeal; notify AI Governance Council	AI provides case summary and supporting SHAP evidence	Human credit analyst makes FINAL determination on ALL appeals	5 business days
Bias Alert Response	Receive alert from monitoring system; notify CRO	AI produces demographic parity report automatically	Human (Ethics Lead + CRO) decides remediation action	24 hours
Model Retirement Decision	N/A	AI drift monitoring flags retirement trigger threshold	Human (ARB + AI Governance Council) approves retirement timeline	30-day process

HITL Design Principles	1. Every consequential AI decision has a human escalation path. 2. Confidence thresholds are set conservatively (err toward HITL). 3. HITL queue capacity is resourced before deployment. 4. HITL effectiveness is monitored monthly.
Approval Authority	HATB-001 approved by: Dr. Amara Diallo (Ethics Lead), James Hartley (CRO), AI Governance Council — 1 March 2026.

AI Extension

AI Operating Model

Owner: Business Architect + HR | ADM Phase: B | TOGAF 10 + AI-First Extension

Organisational Unit	Type	Responsibilities	Operating Cadence	Accountable Owner
AI Governance Council	Cross-functional steering body	Set AI policy; approve high-risk deployments; review AGPs; manage regulatory relationships	Monthly meetings + emergency sessions	James Hartley (CRO, Chair)
Architecture Review Board	EA governance body	Approve AI architecture decisions; enforce standards; review ADRs; conduct compliance assessments	Bi-weekly + ad hoc gate reviews	Dr. Priya Mehta (Chief Architect, Chair)
AI Centre of Excellence (CoE)	Central AI delivery team	Deliver AIDLC; operate MLOps platform; publish AI standards; coach business teams	Permanent team of 12 FTE	Dr. Nina Kowalski (Head of AI CoE)
Domain AI Squads	Embedded AI teams in business units	Execute AI use cases within domain; own business data products; operate production AI systems	Permanent; 2–4 FTE per domain	Domain VPs + AI CoE matrix reporting

Model Risk Management	Risk oversight function	Independent model validation; approve TIER 2 models; conduct ongoing model governance audits	Ad hoc (per model) + quarterly review	CRO direct report
Ethics & Fairness Function	Independent advisory function	Conduct FRIAs; lead fairness evaluations; advise on AGP implementation; manage external ethics board	Per-use-case + ongoing monitoring	Dr. Amara Diallo (Ethics Lead)
MLOps Platform Team	Platform engineering function	Operate MLflow, Azure ML, LLM Gateway, vector database, monitoring stack for all AI systems	24/7 production support + continuous improvement	Priya Patel (MLOps Lead)

TOGAF ADM
Phase C1

Data Architecture

ARTIFACTS: DAB-001 - DM-001 - VDA-001 - FLS-001

PHASE OUTPUT

Data Architecture Blueprint

Owner: Data Architect | ADM Phase: C1 | TOGAF 10 + AI-First Extension

Data Architecture Pattern	Data Lakehouse (Apache Iceberg on Azure Data Lake Gen2) + Data Mesh domain ownership + Feature Store for ML serving consistency
Primary Platform	Azure Data Lake Gen2 (storage) + Azure Databricks (Iceberg/Delta) + dbt (transformations) + Azure Event Hubs (streaming)
Data Domains	6 domains: Credit & Risk, Customer, Operations, Marketing, HR, Enterprise (shared reference data). Each domain owns its AI training data and RAG knowledge base.
AI Data Products	Each domain publishes: (1) Training datasets (DVC-versioned, quality-certified). (2) Feature Store features (Feast). (3) RAG knowledge base (domain-specific vector namespace in Weaviate).
◆ Lineage Standard	OpenLineage (Apache Airflow + dbt plugins). All data transformations emit lineage events. Lineage graph stored in Apache Atlas. Queryable via Atlan data catalog. EU AI Act Article 10 evidence package auto-generated.
◆ Quality Gates	Data products must achieve quality score ≥90% (completeness, accuracy, freshness, consistency) before certification for AI training use. Quality scores published in Atlan catalog.
◆ PII Handling	All PII fields tagged in Atlan. PII masking enforced at Feature Store and RAG retrieval layers via Microsoft Presidio. No PII may flow into LLM prompts unmasked.
◆ Freshness SLAs	Training data refresh: weekly. Feature Store: 15-minute maximum lag. RAG knowledge bases: 4-hour maximum lag. Inference logs: real-time. Monitoring dashboards: 5-minute lag.

◆ AI EXTENSION

Data Mesh Domain Design

Owner: Data Architect + Domain Architects | ADM Phase: C1 | TOGAF 10 + AI-First Extension

Domain	Key Data Assets	Business Owner / Data Owner	AI Data Products Published	Storage Location	Quality Status
Credit & Risk	Credit Applications, Scoring History, Default Records, Alternative Data (Utility/Rent)	Sarah Chen / Dr. Nina Kowalski	CreditRisk-XGB v1.0 training set; Adverse Action RAG knowledge base; Credit features in Feast	S3://credit-domain/ + Feast:credit-features + Weaviate:credit-kb	98.7% completeness; lineage 100% coverage
Customer	CRM records, Interaction History, Segment Data, Consent Records	Lena Hoffmann / Domain Data Owner	CSCopilot RAG knowledge base; Customer segment features	S3://customer-domain/ + Weaviate:customer-kb	97.2%; GDPR consent tracking automated

Operations	Ticket Data, Process Logs, SLA Records, Agent Interaction Logs	Ops VP / Domain Data Owner	Operations knowledge base for Copilot; Process optimisation features	S3://ops-domain / + Weaviate:ops-kb	94.1%; freshness SLA 2h
Security	Transaction Logs, Fraud Labels, Anomaly Patterns, Network Logs	John Kim / Domain Data Owner	FraudDet-ENS v3.1 training; real-time feature feed	S3://security-domain/ + Feast:fraud-features (real-time)	99.3%; p99 freshness <5min (critical)
Marketing	Campaign Data, Response Rates, Channel Attribution, Consent	Marketing VP / Domain Data Owner	Campaign Targeting AI training set; segment propensity features	S3://marketing-domain/ + Feast:mkt-features	95.8%; GDPR consent flag mandatory
HR	Job Postings, CV Pool (anonymised), Assessment Data, Outcome Labels	Maria Santos / Domain Data Owner	CV Screening AI (ON HOLD — FRIA pending); Workforce planning features	S3://hr-domain/ (restricted access)	RESTRICTED ; FRIA must complete before AI use

◆ AI EXTENSION

Vector Database Architecture

Owner: Data Architect + AI Architect | ADM Phase: C1 | TOGAF 10 + AI-First Extension

Platform Selected	<i>Weaviate (self-hosted on AKS) — selected over Pinecone (lock-in risk) and Azure AI Search (limited OSS portability). See ADR: EA-C1-VDA-001.</i>
Embedding Models	<i>text-embedding-3-large (OpenAI) for knowledge bases; domain-specific fine-tuned E5-large for credit/regulatory content. All models versioned in MLflow.</i>
Namespace Architecture	<i>One Weaviate namespace per domain-use-case: credit-kb, customer-kb, ops-kb, regulatory-kb (shared), code-kb (IT). Namespace isolation enforces data product boundaries.</i>
◆ Access Control	<i>Weaviate API key per consuming service. Agent access requires scoped API key registered in HashiCorp Vault. No cross-namespace access without ARB approval.</i>
◆ Freshness SLA	<i>credit-kb: ≤4 hours. regulatory-kb: ≤1 hour (regulatory changes are time-sensitive). All others: ≤8 hours. Freshness monitored in Datadog; breach triggers P2 alert.</i>
◆ Embedding Drift Monitoring	<i>Weekly cosine similarity baseline check across namespaces. >10% average cosine drift triggers embedding model review and potential re-embedding. Monitored via Arize AI.</i>
◆ GDPR Deletion Capability	<i>Weaviate supports object-level deletion by UUID. DPO-triggered PII deletion workflows documented and tested. 72-hour deletion SLA from subject access request.</i>
◆ EU AI Act Evidencing	<i>Embedding model versions stored in MLflow with training data provenance. Namespace contents versioned with DVC. Full lineage from source document to vector queryable via OpenLineage.</i>

TOGAF ADM
Phase C2

Application Architecture

ARTIFACTS: LLMG-001 - AOB-001 - CSD-001 - ASAP-001

◆ AI EXTENSION

LLM Gateway Architecture

Owner: AI Architect + Security Architect | ADM Phase: C2 | TOGAF 10 + AI-First Extension

◆ **AI-First Extension: The LLM Gateway is the mandatory control plane for ALL LLM API calls within GlobalBank. No LLM call may bypass the gateway. This is the application-layer equivalent of the network security perimeter for AI traffic.**

Platform	Kong AI Gateway (enterprise licence) — deployed on AKS in GlobalBank's Azure subscription
Model Routing	LiteLLM abstraction layer behind Kong. Routes to: Azure OpenAI (primary), Anthropic Claude via Bedrock (secondary), Llama 3 (self-hosted, sensitive data). Failover automatic.
◆ Mandatory Controls	Every LLM call through gateway must pass: (1) Prompt injection detection (Lakera Guard API). (2) PII masking (Presidio). (3) Content safety filter (Azure AI Content Safety). (4) Token budget enforcement. (5) Audit log write.
◆ Rate Limiting	Per-service rate limits: Credit scoring LLM = 500 RPM. Customer Copilot = 2000 RPM. Developer Copilot = 5000 RPM. Burst: 2x sustained for 60s. Exceeding burst → 429 + alert.
◆ Audit Logging	100% of LLM requests logged to Splunk: service ID, model, token count, latency, safety score, redacted prompt summary. PII-free log. Immutable append-only. Retained 7 years (FCA).
◆ Cost Attribution	Every LLM call tagged with: cost-centre, use-case-ID, model-version, environment. FinOps dashboard in Azure Cost Management. Weekly cost report to business owners.
◆ Response Caching	Semantic caching enabled for deterministic queries (e.g., regulatory Q&A;). Cache TTL = 4 hours. Cache hit rate target >30% to reduce costs. Privacy: cached responses never contain PII.

◆ AI EXTENSION

Agent Orchestration Blueprint

Owner: AI Architect | ADM Phase: C2 | TOGAF 10 + AI-First Extension

Orchestration Framework	LangGraph (primary for stateful multi-step agents) + LangChain (for RAG chains). AutoGen reserved for experimental multi-agent workflows pending governance maturity.
Integration Protocol	Model Context Protocol (MCP) as the standard for agent-to-enterprise-service integration. All enterprise APIs exposed via MCP servers to agents. No direct REST calls from agents.
◆ Agent Identity Model	Each agent has a dedicated service identity in Azure Managed Identity. Agent identity is SEPARATE from human IAM. Agent credentials are short-lived (15-min TTL). Stored in HashiCorp Vault.
◆ Action Boundaries	Agent action boundaries defined in Agent Action Boundary Register (AABR-001-EA). Enforced at runtime by OPA (Open Policy Agent) sidecar on each agent pod. Violations logged and alerted.

◆ ATF Trust Levels	Agents are assigned ATF trust levels 0–4 by the AI Governance Council. Promotion requires: demonstrated accuracy over evaluation period, security audit, clean operational history, explicit stakeholder approval.
◆ Memory Architecture	Working memory: in-context (per session, ephemeral). Episodic memory: Redis (session history, 24h TTL). Semantic memory: Weaviate vector store (persistent, access-controlled). No unlimited persistent memory.
◆ Circuit Breakers	All agents wrapped with Resilience4j circuit breaker. If error rate >5% in 60s window → circuit opens, agent paused, P1 alert. Prevents runaway agent cascades.

◆ AI Extension

Agent-Safe API Design Pattern

Owner: Solution Architect | ADM Phase: C2 | TOGAF 10 + AI-First Extension

Design Requirement	Standard	Example Implementation
Idempotency	All state-changing endpoints MUST be idempotent (safe to retry). Idempotency key in header.	POST /decisions: X-Idempotency-Key: {uuid} prevents duplicate credit decisions on agent retry
Semantic Versioning in Metadata	API version communicated in response metadata, not just URL. Allows agents to self-discover deprecation.	Response header: X-API-Version: 2.3.1, X-API-Deprecation-Date: 2027-01-01
Action Documentation	Every state-changing endpoint documents: action type (reversible/irreversible), scope of impact, human approval requirement.	POST /decisions: {"reversible": false, "scope": "customer-credit-file", "hitl_required": true}
Burst Rate Limits	Agent-specific rate limits documented in API spec. Differentiated from human user limits. Exponential backoff required.	Agent client: burst 500 RPM sustained 200 RPM. Retry-After header provided on 429.
Webhook Callbacks	Long-running operations return 202 + callback URL. Agents register webhook endpoint. No synchronous waiting.	POST /model-retraining → 202 {callbackUrl: "/retrain-status/{job-id}"}
Dry-Run Mode	Irreversible operations support ?dryRun=true parameter for agent testing without side effects.	DELETE /model-version?dryRun=true → returns what would happen, no deletion
Audit Trail in Response	Every state-changing response includes audit trail ID for traceability from agent action to business outcome.	Response: {"auditId": "AUD-2026-FIN-048371", "agentId": "AGT-OPS-001"}

TOGAF ADM
Phase D

Technology Architecture

ARTIFACTS: AIB-001 - MPD-001 - ZTA-001 - FAOD-001

PHASE OUTPUT

AI Infrastructure Blueprint

Owner: Technology Architect + MLOps Lead | ADM Phase: D | TOGAF 10 + AI-First Extension

Component	Platform	Purpose	Scale Design	Governance Notes
GPU Compute (Training)	Azure NC-series (A100 80GB) — on-demand	Model training, fine-tuning, large-scale batch inference	Spot instances for training; reserved for inference	FinOps: per-use-case cost tagging; weekly budget review
GPU Compute (Inference)	Azure NP-series (inference-optimized)	Real-time model serving via KServe	Autoscaling 2–20 replicas; HPA on GPU util	p99 latency SLA <200ms for credit scoring
Inference Serving	KServe on AKS (v0.13)	Serve XGBoost, PyTorch, ONNX models	Canary deployments; shadow mode; A/B traffic splitting	Model version rollback <3min
Model Registry	MLflow (Azure ML workspace)	Versioned model artifacts, metrics, lineage, deployment history	Git-like versioning; SHA-256 model hash verification	EU AI Act Art.11 technical documentation auto-generated
Feature Store	Feast (self-hosted on AKS + Redis + Azure SQL)	Real-time and batch feature serving; training-serving consistency	Redis for online serving (<5ms); Parquet for offline training	Training-serving skew monitoring via Arize AI
Vector Database	Weaviate (AKS, 3-node cluster)	Semantic search, RAG retrieval, embedding storage	Horizontal scaling; namespace-level ACL	4-hour freshness SLA; embedding drift monitoring
LLM Gateway	Kong AI Gateway (AKS)	Centralised LLM traffic control, safety, audit	Active-active HA; 99.9% SLA	Zero single-point-of-failure; circuit breaker
Observability	Datadog (APM + ML monitoring) + Arize AI (model monitoring) + Splunk (SIEM)	Full-stack observability: infra, application, ML model behaviour, security events	15-second metric resolution; 1-year retention	PagerDuty integration; escalation to on-call MLOps
CI/CD / MLOps Pipeline	Azure DevOps + GitHub Actions + MLflow + dbt	Automated model training, validation, deployment; data pipeline orchestration	4-eyes principle: automated gate + human approval at Phase 6	Full pipeline execution <2 hours for standard models

AI EXTENSION

Zero Trust AI Security Architecture

Owner: Security Architect | ADM Phase: D | TOGAF 10 + AI-First Extension

♦ **AI-First Extension: Traditional Zero Trust assumes human identity. This document extends ZTA to cover AI agent identity, agentic action boundaries, model supply chain security, and AI-specific threat mitigations.**

ZTA Principle for AI	Implementation Detail	Technology Controls
Never Trust Agent Identity	Each agent has a unique, short-lived (15-min TTL) managed identity. No agent inherits human user permissions. Agent identity re-verified at every tool call, not just session initiation.	Azure Managed Identity + HashiCorp Vault dynamic secrets + OPA policy engine
Least-Privilege Action Boundaries	Agents granted minimum tool access for defined task only. Tool access scoped to session, not persistent. Privilege escalation auto-triggers human review + P1 alert.	OPA sidecar enforcement + Agent Action Boundary Register (AABR-001-EA) as policy source
Assume Breach at Agent Layer	Security design assumes any agent can be compromised via prompt injection or jailbreak. Circuit breakers limit blast radius. All agent actions reversible where possible.	Lakera Guard + Azure Content Safety + Resilience4j circuit breakers + dead-letter queues
Model Supply Chain Verification	All model artifacts verified against SHA-256 hash in MLflow before deployment. SBOM (Software Bill of Materials) maintained for all model dependencies. Third-party models undergo TPRM assessment.	Sigstore model signing + MLflow hash verification + Snyk dependency scanning + TPRM vendor questionnaire
Zero Trust Data Access	RAG retrieval respects data namespace ACLs. Agents cannot access data outside their registered namespace. PII masking applied before any LLM processing.	Weaviate namespace ACLs + Presidio PII masking + Azure AD Conditional Access for data layer
Continuous Verification	Security posture of deployed agents re-assessed quarterly. Anomaly detection on agent behaviour patterns flags deviation from baseline.	Splunk UBA (User and Entity Behaviour Analytics) extended to cover agent identities; quarterly security audit
Audit Immutability	All agent actions logged to immutable, tamper-evident audit log (WORM storage). Log integrity verified via cryptographic hash chain. Accessible to FCA on request.	Azure Blob WORM storage + SHA-256 hash chain + Splunk SIEM; 7-year retention

TOGAF ADM
Phase EF

Opportunities & Migration

ARTIFACTS: AUP-001 - ARTA-001 - ARM-001

AI EXTENSION

AI Use Case Portfolio & Sequencing Plan

Owner: Chief Architect + AI CoE Lead | ADM Phase: EF | TOGAF 10 + AI-First Extension

AI Use Case	Business Value	Strategic Fit	Risk Tier	Est. Duration	Current Status	Next Milestone	Target Live	Key Constraints
Credit Risk AI Scoring	£7.3M p.a.	HIGH	T2 HIGH	14 months	ARB Approved Q4 2025	Q1 2026 Phase 3	Q3 2026 Phase 7	Foundation use case; FRIA complete; external audit booked
Fraud Detection Governance Uplift	£2.1M (risk reduction)	HIGH	T2 HIGH	3 months	Production already — governance gap	Q1 2026 Governance Audit	Q2 2026 Complete	Existing system; compliance uplift only; fast track
Customer Service Copilot Expansion	£1.4M incremental	MEDIUM	T3 LIMITED	6 months	Live v2.1 — expand to more channels	Q2 2026 Scope Extension	Q3 2026 Full deployment	Low risk; accelerated AIDLC track
Campaign Targeting AI	£0.9M p.a.	MEDIUM	T4 MINIMAL	9 months (in progress)	Phase 5 Development	Already in progress	Q3 2026 Phase 7	AIDLC-Lite track; consent management focus
Developer Copilot (GitHub Copilot)	Productivity — unmeasured	MEDIUM	T4 MINIMAL	1 month (register only)	Shadow AI — in use	Immediate registration	Done — ongoing monitoring	Register in ASI-001; code audit policy; AIDLC-Lite
CV Screening Assistant	£0.6M (efficiency)	LOW	T2 HIGH	18+ months	ON HOLD — FRIA required	FRIA Q2 2026	Subject to FRIA	EU AI Act Annex III; highest scrutiny; FRIA outcome gating

Sequencing Rationale	Risk-adjusted sequencing: (1) High business value + compliance foundation first (Credit Risk). (2) Production gap-fills second (Fraud governance). (3) Low-risk progressions third. (4) High-scrutiny on hold until governance matures.
Architecture Dependencies	Credit Risk AI (Phase 7) is the REFERENCE IMPLEMENTATION. MLOps platform, LLM Gateway, vector database, and monitoring stack must be operational before any subsequent AI system can reach Phase 7.

AI EXTENSION

AI-Readiness Transition Architecture

Owner: Chief Architect + Solution Architects | ADM Phase: EF | TOGAF 10 + AI-First Extension

Architecture State	Data Architecture	MLOps / LLMOps	Application Architecture	Agent Architecture	Security Architecture	Governance	AI Maturity Level
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Baseline (Q1 2026)	Data Lake (unstructured, no lineage)	No MLOps; manual training+deploy	LLM calls via direct API (no gateway)	No agent infrastructure	Perimeter only; no ZTA	Ad hoc; no ARB process	L1 (Initial) — McKinsey AI Maturity
Transition 1 (Q2 2026)	Lakehouse + DVC versioning; 2 domains on Mesh; OpenLineage installed	MLflow model registry; Azure ML pipelines; Feast MVP	Kong LLM Gateway live; Lakera Guard + Presidio active	LangGraph framework standard; OPA agent policy MVP	Agent identity registry; HashiCorp Vault deployed	ARB gates for AI systems; AGP published	L2 (Developing)
Transition 2 (Q3 2026)	Full Data Mesh (6 domains); Weaviate production; Feature Store certified	Full LLM Ops monitoring: Arize AI, hallucination detection, drift alerts	RAG pipelines production-grade; MCP standard adopted for new APIs	3 production agents; ATF trust levels active; circuit breakers live	Zero Trust for agents fully implemented; MAESTRO threat model done	ISO/IEC 42001 gap assessment; EU AI Act compliance audit	L3 (Scaling)
Target State (Q4 2026)	Enterprise AI Data Catalog (Atlan); automated lineage; all AI data products quality-certified	Autonomous retraining MVP; unified MLOps+LLMOps platform	Agent mesh production; full MCP integration; semantic caching	Agentic workflows in production across 3 business domains	Full MAESTRO controls; quarterly zero trust review	ISO/IEC 42001 certified; EU AI Act high-risk documented	L4 (Optimizing)

TOGAF ADM
Phase G

Implementation Governance

ARTIFACTS: ACC-001 - ASRF-001 - ARB-001

PHASE OUTPUT

Architecture Compliance Checklist

Owner: Architecture Review Board | ADM Phase: G | TOGAF 10 + AI-First Extension

PURPOSE: Every AI system must complete this checklist before the ARB grants Architecture Compliance Certificate (ACC). This checklist integrates TOGAF Phase G governance with AIDLC Phase 6 (Evaluation) gate requirements.

Compliance Artefact	AIDLC Phase	Requirement	Evidence Provided	Status
Use Case Charter (AUC-001)	AIDLC Phase 1	AI use case formally defined with business value hypothesis, risk flags, and sponsor sign-off	AUC-2026-FIN-047 v1.2 — Approved 19 Feb 2026	PASS
Risk Classification (RCS-001)	AIDLC Phase 1	EU AI Act risk tier assigned; governance intensity set	RCS-001 — T2 HIGH RISK — 22 Feb 2026	PASS
Feasibility Report (FR-001)	AIDLC Phase 2	Technical and organisational feasibility confirmed	FR-001 v1.0 — 5 March 2026	PASS
FRIA Completed (FRIA-001)	AIDLC Phase 2	Fundamental Rights Impact Assessment completed for T2 systems	FRIA-001 — Signed CRO + Ethics Lead 10 Mar 2026	PASS
Architecture Decision Records (ADR)	AIDLC Phase 4	All key architecture decisions documented with options and rationale	ADR-2026-FIN-047-001 — ARB Approved 2 Apr 2026	PASS
Constitutional AI Policy (CAP)	AIDLC Phase 4	Constitutional AI Policy approved by AI Governance Council	CAP-001 v1.0 — AI Gov Council 5 Apr 2026	PASS
Red Team Report (RTR-001)	AIDLC Phase 6	All HIGH severity findings resolved; MEDIUM findings remediated or accepted	RTR-001 — All HIGH resolved 22 Apr 2026	PASS
Fairness Evaluation (FER-001)	AIDLC Phase 6	All fairness metrics within thresholds; external audit completed for T2	FER-001 — AJI External Audit 22 Apr 2026	PASS
Constitutional Compliance Audit (CCA)	AIDLC Phase 6	All 8 Constitutional AI Principles assessed and PASS	CCA-001 — 100% compliance 25 Apr 2026	PASS

AI System Inventory Registration	EA Cross-cutting	System registered in ASI-001 with complete metadata	ASI-001 — AUC-2026-FIN-047 registered	PASS
Agent Action Boundary Register	EA Cross-cutting	Agent boundaries documented for any agentic components	AABR-001-EA — Non-agent system; N/A	N/A
Data Lineage Map (DLM-001)	AIDLC Phase 3	End-to-end data lineage documented and accessible via Atlan	DLM-001 — All 7 nodes documented	PASS
Deployment Runbook (DRB-001)	AIDLC Phase 7	Blue/green deployment plan with rollback procedures documented	DRB-001 — MLOps Lead sign-off 30 Apr 2026	PASS
User Disclosure Documentation	AIDLC Phase 7	Customer-facing AI disclosure text legal-reviewed and approved	UDD-001 — Legal sign-off 2 May 2026	PASS

ARB Decision	ARCHITECTURE COMPLIANCE CERTIFICATE GRANTED — CreditRisk-XGB-v1.0. Authorised for deployment. Approval: Dr. Priya Mehta (Chief Architect), James Hartley (CRO), AI Governance Council — 5 May 2026.
Conditions of Certificate	1. Monthly Monitoring Reports (MMR-001) reviewed by ARB quarterly. 2. Annual external fairness audit maintained. 3. Any material model change requires new ACC application.

✦ AI EXTENSION

AI System Registration Form

Owner: Business Owner + AI CoE | ADM Phase: G | TOGAF 10 + AI-First Extension

System ID (assigned by AI CoE)	AUC-2026-FIN-047
System Name	Credit Risk AI Scoring Model
Business Owner	Sarah Chen, VP Credit Risk, Retail Banking Division
Technical Owner	Dr. Nina Kowalski, Head of AI CoE
System Description	ML-based credit scoring model (XGBoost) with LLM adverse action notice generator (GPT-4o fine-tuned). Used for automated credit decisions on UK retail lending applications (50,000/month).
AI Category	ML Classification (primary) + Generative AI (adverse action narrative)
EU AI Act Risk Tier	TIER 2 — HIGH RISK (Annex III, Category 5b: creditworthiness assessment)
Regulated By	FCA, PRA, ICO, EU AI Act National Competent Authority
Personal Data Processed	YES — Applicant financial history, utility payment history, rent history. Processing basis: Art. 6(1)(b) GDPR (contractual necessity).
Third-Party AI Components	Equifax Credit Bureau API, GPT-4o (Azure OpenAI), Lakera Guard, Microsoft Presidio
Deployment Environment	Production — Azure Kubernetes Service (AKS), UK South region
Date Registered	5 May 2026

ACC Certificate Number	ACC-2026-FIN-047-001 (Architecture Compliance Certificate granted 5 May 2026)
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TOGAF ADM
Phase H

Architecture Change Mgmt

ARTIFACTS: ACR-001 - APR-001 - AIM-001

PHASE OUTPUT

Architecture Change Request

Owner: Requestor + Architecture Review Board | ADM Phase: H | TOGAF 10 + AI-First Extension

Change Request ID	ACR-2026-FIN-047-002
Date Submitted	15 June 2026
Requestor	Dr. Nina Kowalski, Head of AI CoE
System Affected	CreditRisk-XGB-v1.0 (AUC-2026-FIN-047)
Change Title	Integrate Real-Time Rental Bureau Data (Experian API) — Model Retraining
Change Description	Add Experian Rental Bureau as a new alternative data source for credit scoring. Requires: (1) New data pipeline (rental-bureau-pipeline). (2) Feature engineering for 6 new rental features. (3) Model retraining with new features. (4) Fairness re-evaluation (new data source may affect demographic groups). (5) Red team re-assessment of new data pipeline.
Change Classification	SIGNIFICANT — New data source changes model behaviour and requires Phase 5-6 re-run for new features. Does NOT require full Phase 1-4 restart.
◆ AIDLC Impact Assessment	Phases required: Phase 3 (Data Sheet update for rental data) → Phase 5 (Retrain with new features) → Phase 6 (Fairness re-evaluation, bias re-assessment) → Phase 7 (Re-deployment runbook). Estimated: 6 weeks.
◆ Regulatory Notification Required	NO — change is additive, does not alter the fundamental model architecture or purpose. Legal review confirms no new regulatory obligations.
Architecture Risk Assessment	LOW-MEDIUM: (1) New data source may reveal unexpected bias (mitigated: mandatory re-run of FER-001). (2) Vendor reliability of Experian Rental Bureau API (mitigated: fallback to bureau-only scoring if API down).
ARB Decision	APPROVED with conditions: (1) FER-001 re-run must show no degradation in fairness metrics. (2) Rental data DPA signed before data ingestion. (3) Updated Data Sheet signed by Data Governance Board. — ARB 22 June 2026

◆ AI EXTENSION

AI Portfolio Review Report

Owner: Chief Architect + AI CoE Lead | ADM Phase: H | TOGAF 10 + AI-First Extension

◆ AI-First Extension: Quarterly review of the full AI system portfolio. Aggregates monitoring data from all deployed systems, identifies patterns and risks, and makes portfolio-level architectural recommendations.

Review Period	Q2 2026 (April–June 2026)						
Portfolio Status	5 systems in production, 1 in development (Campaign Targeting), 1 on hold (CV Screening)						
System	Tier	Status	RA G	Model Performance	Fairness Status	Incidents	Action Required

CreditRisk-XGB-v1.0	T2	Production	GREEN	0.827 AUC (↓0.3% from baseline; within threshold)	1.7pp (threshold: 2.0pp) — WATCH	0 P0, 0 P1, 1 P2 resolved	Rental data integration approved (ACR-002); no immediate action
CSCopilot-v2.1	T3	Production	GREEN	4.3/5 CSAT (↑0.2 from Q1)	N/A (non-credit; no parity requirement)	0 P0, 0 P1, 2 P3 resolved	Expand to mobile channel Q3; backlog prioritised
FraudDet-ENS-v3.1	T2	Production	AMBER	AUROC 0.94 (stable); TPR 91% (↓2% from Q1)	N/A (anomaly detection)	1 P1 (false positive spike wk 4; resolved) — learning point	Investigate TPR degradation; retraining evaluation in progress
DevCopilot (GitHub Copilot)	T4	Production	GREEN	Developer satisfaction 4.6/5; code review escalations: 3/month	N/A	0 incidents	AIDLC-Lite compliance verified; shadow AI register clean
CampaignTarget-v0.3	T4	Development	N/A	In Phase 5 development; no production metrics	N/A	N/A	Phase 6 evaluation scheduled Q3; consent management integration on track
CVScreening (HOLD)	T2	On Hold	N/A	Not deployed; FRIA in progress	N/A	N/A	FRIA expected Q3 2026; proceed only if FRIA clears

Portfolio-Level Risks	1. FraudDet TPR degradation (AMBER) — prioritise retraining evaluation. 2. Credit Risk bias delta approaching threshold (WATCH) — increase monitoring frequency.
Architecture Recommendations	1. Fraud Detection model retraining approved — execute Phase 5 re-run by end Q3. 2. Strengthen fairness monitoring alerting: reduce bias delta alert threshold from 2.0pp to 1.8pp across all T2 systems.
Next Portfolio Review	Q3 2026 — 15 September 2026

TOGAF ADM
Phase EA

EA Cross-Cutting Artifacts

ARTIFACTS: EADR-001 - AABR-EA-001 - AS-EA-001 - RAIMP-001

EA CROSS-CUTTING

EA Architecture Decision Record (AI-First)

Owner: Architecture Review Board | ADM Phase: EA | TOGAF 10 + AI-First Extension

PURPOSE: Documents enterprise-level architecture decisions that affect multiple AI systems or the AI platform itself. Distinct from AIDLC Phase 4 ADRs (which are use-case-specific). EA ADRs are binding on all teams and permanent record.

EA ADR ID	EADR-2026-PLAT-003
Title	LLM Provider Strategy: Multi-Provider via LiteLLM Router — Mandatory Standard
Status	ACCEPTED — Architecture Review Board — 15 March 2026
Decision Makers	Dr. Priya Mehta (Chief Architect), AI Governance Council, Technology Leadership
Context	GlobalBank has three AI systems consuming LLM APIs from three different providers (Azure OpenAI, Anthropic, self-hosted Llama 3). Without standardisation, each team manages its own client library, rate limiting, cost tracking, safety filtering, and audit logging. This creates governance gaps, duplicated effort, and single-provider dependency risk.
Decision	ALL LLM API calls from GlobalBank systems MUST be routed through the Kong AI Gateway with LiteLLM abstraction layer. Direct LLM provider API calls are prohibited. Providers: Azure OpenAI (primary), Anthropic Claude (secondary via Bedrock), Llama 3 (self-hosted, sensitive data). Failover automated by LiteLLM.
Options Considered	(A) Multi-provider via LiteLLM gateway (SELECTED). (B) Azure OpenAI only (rejected: vendor lock-in risk). (C) Self-hosted LLMs only (rejected: capability gap). (D) Each team manages own clients (rejected: governance impossible).
Consequences (Positive)	Single governance point for LLM traffic. Unified audit log. Cost attribution. Safety filtering standardised. Provider switch without code changes. 30-40% cost reduction via semantic caching.
Consequences (Negative)	Gateway becomes single point of failure (mitigated: active-active HA). Migration effort for 3 existing systems (estimated: 2 sprints each). LiteLLM abstraction adds ~5ms latency (acceptable).
Compliance Link	EU AI Act Article 12 (logging); NIST AI RMF GOVERN-1.7 (documentation); GlobalBank AGP-6 (security); AGP-7 (accountability)
Review Date	Reviewed annually or when major provider change occurs

EA CROSS-CUTTING

Enterprise Agent Action Boundary Register

Owner: AI Architect + Security Architect | ADM Phase: EA | TOGAF 10 + AI-First Extension

Register Purpose	Single source of truth for all AI agent action boundaries. Enforced at runtime by OPA sidecar. Updated when agents are deployed or their scope changes. Reviewed quarterly by ARB.
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Enforcement Mechanism		OPA (Open Policy Agent) reads AABR as policy source. Agent presents signed manifest with claimed identity. OPA validates identity + requested action against AABR before execution.					
Agent ID	System	Framework	Permitted Actions	Denied Actions	ATF Level / HITL	Hard Boundaries	Owner
AGT-FIN-001	CreditRisk-XGB-v1.0	Scorer (non-agent)	bureau-api:READ, utility-api:READ, temenos-api:READ, decision-log:WRITE	ALL payment APIs, account-modify-api, customer-data-api:WRITE	ATF 1 — All decisions confirmed by core banking	None — non-agent	CRO
AGT-OPS-001	CSCopilot-v2.1	LangGraph stateful	crm-api:READ, kb-search:READ, ticket-api:CREATE-ONLY, order-api:READ	payment-api:ALL, account-modify-api:ALL, pii-export:ALL	ATF 3 — Notify human; refunds >£100 need HITL	No payment; no account mod; no PII export	VP Operations
AGT-IT-001	InfraAgent-v1.0	AutoGen sub-agent	cloudwatch:READ, github-api:PR-CREATE-ONLY, slack-api:POST	production-deploy:ALL, iam:ALL, secrets:ALL, db:WRITE	ATF 2 — All code changes need human PR approval	No prod deploys; no IAM; no secret access	CISO
AGT-DATA-001	DataQuality Agent-v1.0	LangChain single-agent	data-catalog:READ, quality-metrics:READ, atlan-api:TAG-WRITE	training-data:WRITE, feature-store:WRITE, s3:DELETE	ATF 2 — Quality remediations >Tier2 need approval	No training data modification; no feature deletion	Chief Data Officer

EA CROSS-CUTTING

Responsible AI Maturity Programme

Owner: AI Governance Council | ADM Phase: EA | TOGAF 10 + AI-First Extension

Maturity Level	Characteristics	Remaining Gaps	GlobalBank Timeline
L1 — Initial	Ad hoc AI deployments; no governance; reactive to incidents	Shadow AI; no bias testing; no HITL; no audit trail; regulatory non-compliant	No programme in place
L2 — Developing	AIDLC framework adopted; AI Governance Council established; AGPs published; first ARB gate process	Some systems lack full compliance; MLOps immature; data lineage partial; no zero trust	Q1 2026 — GlobalBank baseline
L3 — Scaling (TARGET 2026)	Full AIDLC for all T2 systems; LLM Gateway live; Data Mesh 6 domains; MLOps+LLMOps unified platform; ZTA for agents; ISO/IEC 42001 gap closed	Some T3/T4 systems still on AIDLC-Lite; autonomous retraining not yet live	Q4 2026 — Target state
L4 — Optimising	Continuous governance monitoring; autonomous retraining; AI portfolio optimisation; proactive regulatory engagement; industry-leading transparency reporting	Constant vigilance required; risk of complacency	2027 target

L5 — Leading	Industry benchmark; external thought leadership; contributing to standards bodies (The Open Group, NIST, EU AI Act implementation); open-sourcing governance tooling	Resource-intensive to maintain	2028+ aspiration
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Current Maturity Level (May 2026)	L2 transitioning to L3. AI Governance Council operational. AIDLC in use for 3 production systems. MLOps platform MVP live. Data Mesh 2 of 6 domains. Zero Trust scoped but not deployed.
Target by Dec 2026	L3 — Scaling. ISO/IEC 42001 certified. EU AI Act high-risk documentation complete. All 6 Data Mesh domains live. Full ZTA deployed. LLM Gateway enforced enterprise-wide.
KPIs	% AI systems with full AIDLC coverage (target 100% T2); Bias delta all T2 systems <2pp; HITL trigger rate within designed range; Governance cost as % of AI programme (target <15%)

ARTIFACT CROSS-REFERENCE BY ADM PHASE

ADM Phase	Artifact Templates in This Library
PRE — Preliminary	E AFC-001 EA Framework Charter, A GP-001 AI Governance Principles, E SR-001 Stakeholder Register
A — Architecture Vision	S AW-001 Statement of Architecture Work, A CA-001 AI Capability Assessment, A VD-001 Architecture Vision
B — Business Architecture	A CM-001 AI Capability Map, H ATB-001 Human-AI Task Boundary Map, A OM-001 AI Operating Model, W IA-001 Workforce Impact Analysis
C1 — Data Architecture	D AB-001 Data Architecture Blueprint, D M-001 Data Mesh Domain Design, V DA-001 Vector DB Architecture, F LS-001 Feature Store Lineage Spec
C2 — Application Architecture	L LMG-001 LLM Gateway Architecture, A OB-001 Agent Orchestration Blueprint, C SD-001 Copilot Stack Design, A SAP-001 Agent-Safe API Pattern
D — Technology Architecture	A IB-001 AI Infrastructure Blueprint, M PD-001 MLOps Platform Design, Z TA-001 Zero Trust AI Architecture, F AOD-001 FinOps for AI Design
E/F — Opportunities & Migration	A UP-001 AI Use Case Portfolio, A RTA-001 AI-Readiness Transition Architecture, A RM-001 Architecture Roadmap
G — Implementation Governance	A CC-001 Architecture Compliance Checklist, A SRF-001 AI System Registration Form, A RB-001 ARB Decision Log
H — Change Management	A CR-001 Architecture Change Request, A PR-001 AI Portfolio Review Report, A IM-001 Architecture Impact Model
EA — Cross-Cutting	E ADR-001 EA Architecture Decision Record, A ABR-EA-001 Agent Boundary Register, A SI-EA-001 AI System Inventory, R AIMP-001 RAI Maturity Programme

LIBRARY SCOPE: 30+ EA lifecycle artifact templates across all TOGAF 10 ADM phases with AI-First extensions. Standard TOGAF artifacts are extended with ♦ AI-First fields covering: AI governance, Constitutional AI policy, Human-AI task boundaries, agent orchestration, zero trust for AI, and regulatory compliance. All templates shown with example content for the "GlobalBank AI Transformation Programme 2026." Replace blue italic values with your organisation's actual content.